

CFDA at TREC 2026 Retrieval-Augmented Generation Track

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Abstract

We developed separate CFDA pipelines for the Retrieval (R) and Retrieval-Augmented Generation (RAG) tasks of TREC RAG 2026. The Retrieval pipeline combines multiple query routes, protected-head reranking, cross-encoder-guided deep-tail reranking, and variable-depth serialization. The RAG pipeline uses a bounded evidence-acquisition loop followed by dense answer writing, sentence-level verification, and citation ordering. On the 22-topic development set, the selected Retrieval output obtains nDCG@10 of 0.7456, Recall@submitted- k of 0.2343, P@10 of 0.9076, MAP of 0.1432, and MRR of 0.9848. The selected RAG output obtains $V_{\text{strict}} = 0.4656$, $A = 0.5298$, $FS = 97.0\%$, and $NS = 0.1\%$. All effectiveness results in this report come from the development set; organizer test-set effectiveness results were unavailable at the time of writing.

1 Introduction

Long-form narratives pose a different retrieval problem from short keyword queries. A single narrative may ask about several entities, mechanisms, historical periods, and comparisons. A lexical retriever can miss a facet whose vocabulary is absent from the original wording, while unconstrained query expansion can introduce unsupported terms and semantic drift. We therefore keep the original narrative as an anchor and retrieve complementary formulations through separate routes.

Generation introduces a related trade-off. A small evidence set may support fluent writing but omit requested facets. Reading more documents can improve coverage, but it also increases context, model calls, and the chance of unsupported claims or weak citations. Citation verification and nugget coverage measure different properties: a well-supported sentence may omit a requested aspect, while a broad answer may contain citations that do not support every claim.

We study these trade-offs in the 2026 TREC RAG setting, which defines separate Retrieval and Retrieval-Augmented Generation tasks over the ClimbMix collection (TREC RAG Organizers, 2026b,a). We ask three questions. **RQ1:** How does the selected multi-route retrieval configuration compare with original-narrative BM25 while retaining the original narrative as an anchor? **RQ2:** What retrieval-depth profile results from variable serialization, and what coverage behavior is observed under bounded evidence acquisition? **RQ3:** What trade-off emerges between answer coverage and citation support across different stored RAG configurations?

The report describes multi-route, variable-depth Retrieval with a protected head and deep-tail reranking; a bounded evidence-acquisition loop for citation-aware dense writing; and development-set analyses of deep-tail ranking and the coverage-citation-support trade-off across stored RAG configurations. *All numerical results in this paper are from the 22-topic development set unless explicitly stated otherwise.*

2 Related Work

2.1 Query Expansion and Hybrid Retrieval

BM25 provides a strong lexical first-stage retriever for matching term evidence in long narratives (Robertson and Zaragoza, 2009). Query expansion can complement this lexical anchor: Query2Doc uses a large language model to generate a pseudo-document for expansion (Wang et al., 2023), while related work studies prompted query expansion and iterative refinement (Jagerman et al., 2023; Chen et al., 2024b). We use the original narrative and expanded formulations as separate routes, so an expanded query does not replace the lexical anchor.

Hybrid retrieval also combines lexical and semantic signals at different stages. Dense Passage Retrieval learns a dense first-stage representation

(Karpukhin et al., 2020), late interaction retains token-level matching for efficient passage search (Khattab and Zaharia, 2020), and SPLADE makes lexical expansion part of the representation (Formal et al., 2021b,a). Our candidate pool likewise combines complementary retrieval signals; we evaluate their route-level combination and subsequent serialization on the development set.

2.2 Neural Reranking and Rank Fusion

Rank fusion is useful when component rankings provide complementary evidence. Reciprocal Rank Fusion is a simple robust baseline for combining rankings without requiring a common score scale (Cormack et al., 2009). Our Retrieval system applies weighted rank fusion at route-combination stages and again when combining the BM25, cross-encoder, and dense-ranking head signals. A cross-encoder tail rank is then fused with candidate-pool order only over the specified tail, leaving the head and the variable-depth cutoff stages separable. BEIR shows that zero-shot behavior varies substantially across heterogeneous domains and that some neural rerankers can trail BM25 on particular out-of-domain collections (Thakur et al., 2021). This is not a universal weakness of cross-encoders: scaling studies find that sufficiently large cross-encoders can generalize strongly and outperform comparable bi-encoders (Rosa et al., 2022). Our protected-head choice is based on development-set observations for ClimbMix, not on a general claim that cross-encoders degrade out of domain.

Choosing how deep to read a ranked list is itself an open problem separate from how the list is ordered. Early work optimizes a per-query cut-off from the shape of the query’s own score distribution (Arampatzis et al., 2009); more recent work selects a long-context QA read depth from the steepest drop in a sorted similarity-score distribution, without additional tuning or iterative model calls (Taguchi et al., 2025). Our variable-depth serialization (§4.1) is a relative-threshold instance of this same family, applied to the score used immediately before deep-tail reranking rather than to a single-query similarity distribution.

2.3 Evidence-Grounded and Agentic RAG

Retrieval-augmented generation couples evidence selection with answer generation (Lewis et al., 2020). REALM and Atlas demonstrate how retrieval can be integrated with language-model reasoning and few-shot adaptation (Guu et al., 2020;

Izacard et al., 2023); WebGPT similarly studies browser-assisted question answering with feedback (Nakano et al., 2021). Our RAG system also acquires evidence before generation, but it limits additional reading and handles citation ordering in a separate finalization step. Self-RAG addresses when to retrieve by training a single model to emit reflection tokens that decide, and critique, retrieval and generation end to end (Asai et al., 2024); our sufficiency judge makes the analogous retrieve-more-or-stop decision but as an explicit, bounded, inference-time policy rather than a trained-in behavior. The resulting retrieve-or-stop decision is recorded at inference time. IRCOT interleaves retrieval with chain-of-thought steps for multi-step questions (Trivedi et al., 2023). CRAG instead evaluates retrieved evidence and triggers corrective retrieval actions (Yan et al., 2024), while DRAGIN explicitly separates decisions about when to retrieve from how to formulate the next query (Su et al., 2024). Our sufficiency judge combines the retrieve-or-stop and follow-up-query decisions. Unlike these systems, our implementation keeps a fixed generation policy, deterministic query validation, and explicit document and iteration caps.

Evaluation of retrieve/generate systems has increasingly emphasized coverage, attribution, and rubric-based assessment. The TREC RAG workbench and rubric-oriented evaluation studies provide context for our development diagnostics (Dietz, 2024; Farzi and Dietz, 2024; Pradeep et al., 2024, 2025). ALCE evaluates citation-bearing long-form answers along correctness, fluency, and citation-quality dimensions, and shows that apparently strong generated answers can still lack complete citation support (Gao et al., 2023). RAGAS also uses automated judgments (Es et al., 2024), while work on faithfulness evaluation examines their interpretation (Jing et al., 2025). Separately, citation *correctness* (whether a cited passage supports a claim) and citation *faithfulness* (whether the citation reflects genuine grounding rather than post-hoc rationalization) have been shown to diverge substantially, with a large share of formally correct citations found unfaithful under adversarial testing (Wallat et al., 2025); our sentence-level verify/revise stage targets citation correctness. We do not separately measure faithfulness as defined in that study. The reported internal diagnostics are development observations and are not equated with organizer-defined test measures.

3 Task Setup and Evaluation

3.1 Tasks and Output Formats

The 2026 track contains two tasks: Retrieval (R) and Retrieval-Augmented Generation (RAG). The official materials identify ClimbMix as the primary collection and provide separate output conventions for a TREC retrieval run and a JSONL RAG response (TREC RAG Organizers, 2026b,a). Retrieval depth is narrative-specific: a system should submit the documents it predicts to be relevant and useful, without padding every narrative to a conventional fixed cutoff. In our implementation, the R output is serialized as a six-column TREC run, while the RAG output contains one JSON object per narrative with metadata, references, and sentence-level answer objects.

The official RAG contract permits up to 1,024 words per narrative and zero to three citations per answer object. The organizer evaluation is described as anonymized system-by-system battles, individualized nugget rubric scoring in the style of AutoNuggetizer, and weighted citation precision and weighted citation recall (TREC RAG Organizers, 2026a). No organizer-returned test judgments or official test-set scores were available during development.

3.2 Development Set and Metrics

We use a 22-topic development set. For Retrieval, we use three projected UMBRELA qrels (Upadhyay et al., 2024) and a deterministic local evaluator with relevance threshold $\text{rel} \geq 2$. nDCG uses the remaining graded labels as linear gain after labels below 2 are assigned zero gain; P@10, Recall, MAP, and MRR use the corresponding binarized $\text{rel} \geq 2$ judgments. We report nDCG@10, Recall at the depth actually submitted for each topic (Recall@submitted- k), P@10, MAP, and MRR. The fixed top-1000 Recall value is retained only as a fixed-depth comparison; it is not the variable-depth score.

For RAG, V_{strict} is the evaluator’s `strict_vital_score`: the fraction of vital nuggets receiving full support, with partial support counted as zero. A is `all_score`: the coverage over all nuggets, counting full support as 1 and partial support as 0.5. FS, PS, and NS are respectively Full, Partial, and No Support, reported as proportions over judged sentence-citation pairs (with evaluator-error rows excluded). Thus FS and NS are not complementary: the omitted

remainder is PS. These are internal development diagnostics, not organizer-returned official metrics. All effectiveness scores reported below are development results; structural diagnostics from the official 119-topic runs are explicitly labeled as such.

4 Method

4.1 Multi-Route Variable-Depth Retrieval

Figure 1 shows our Multi-Route Variable-Depth Retrieval with Deep-Tail Reranking system. The original narrative remains the primary retrieval signal, with separate budgets for expansion, reranking, and serialization. Candidate membership, ordering, and submitted depth are computed in separate stages.

The system begins with the original narrative. A Query2Doc route generates one pseudo-document and appends it to an anchor-boosted query formed by repeating the original narrative five times, expanding the query without discarding the anchor (Wang et al., 2023). The original-narrative and successful Query2Doc BM25 rankings are fused first (weight 1.0 each; RRF $k = 60$), and the loop reads the first five successfully fetched documents. Only an insufficient judgment elicits follow-up queries. Each accepted query retrieves BM25 to depth 5,000 as a new route of weight 0.25; fusion is recomputed and up to four unseen documents are read. The loop stops on sufficiency, at 12 documents, or after three acquisition iterations; the fused pool contains up to 5,000 documents.

Retained Query2Doc traces use one temperature-zero call, a 400-token limit, and a 180-word pseudo-document cap. A failed or degenerate call omits that route; retrieval continues with the original narrative. Run configurations select `gpt-5.6-sol` for Query2Doc and facets. The `gpt-oss-120b` sufficiency judge instead proposes follow-ups, with up to five attempts per decision. Topic records retain generated Query2Doc and facet content, but no immutable provider snapshot or per-response model identifier.

For a document d , weighted reciprocal rank fusion is

$$s_{\text{RRF}}(d) = \sum_{r: d \in r} \frac{w_r}{k + \text{rank}_r(d)}. \quad (1)$$

Here $k = 60$, the original and Query2Doc routes have weight 1.0, and each accepted follow-up BM25 route has weight 0.25. Equation 1 applies

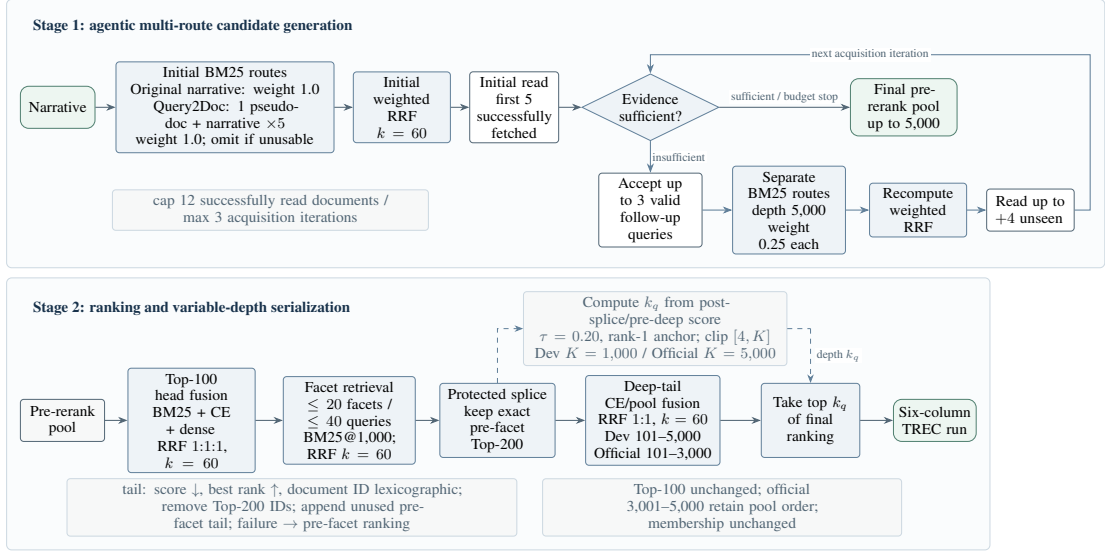


Figure 1: Multi-Route Variable-Depth Retrieval with Deep-Tail Reranking. The diagram depicts the development configuration and separates multi-route acquisition, head reranking, facet-based expansion, deep-tail ordering, and variable-depth serialization. For the official 119-topic run, the deep-tail span was 101–3,000 and the serialization cap was $K = 5,000$, as specified in the text.

to the Retrieval route-fusion stage; the separate top-300 RAG reranker is defined in §4.2.

The fused candidate pool is built to depth 5,000. The top-100 head is then reranked using BM25-rank, MiniLM cross-encoder-rank (Wang et al., 2020), and BGE-M3 dense-embedding-rank (Chen et al., 2024a) signals with equal RRF weights and $k = 60$. The top portion of the pre-facet ranking is protected while facet-based retrieval constructs the complementary tail. Up to 20 schema-validated, searchable facets tied to exact narrative spans each yield one initial query and up to one reformulation that does not use retrieved evidence. After normalized deduplication, each of at most 40 queries retrieves BM25 to depth 1,000. Equal-weight RRF ($k = 60$) accumulates repeated document IDs into a unique pool of up to 5,000. The exact pre-facet top-200 remains unchanged; after its IDs are removed, facet candidates are sorted by descending RRF score, ascending best contributing rank, and then lexicographic document ID. They fill the tail, followed when needed by unused pre-facet tail documents in original order. Failed routes are omitted; if generation or every retrieval fails, the pre-facet ranking is used.

Deep cross-encoder tail reranking. The selected deep-tail stage applies `cross-encoder/ms-marco-MiniLM-L6-v2` below the first 100 entries, which are preserved unchanged. It fuses cross-encoder and candidate-pool tail ranks

with equal RRF weights and $k = 60$. The reranked span is ranks 101–5,000 in development; in the official 119-topic run it is ranks 101–3,000, after which ranks 3,001–5,000 retain candidate-pool order. This stage does not add or remove candidates, but it can change their tail order and which documents fall inside a serialized prefix.

Variable-depth serialization. The topic-specific submission depth is derived from the post-splice, pre-deep-CE candidate-pool score: the implementation specifies a relative threshold of 0.20, a minimum depth of 4, and a maximum depth K that differs between settings: $K = 1,000$ for the development runs and $K = 5,000$ for both official 119-topic submissions. The lower clip is never active in either setting; the smallest depth selected by the threshold is 100 on the development set and 137 on the official topics. Any such cap is a team-imposed limit, as the Retrieval task specification sets no fixed maximum submission depth. The deep-CE stage changes ordering but does not change topic depth. The output is serialized as a six-column TREC run. For topic q , the corresponding serialization rule is

$$k_q = \text{clip}(|\{d : s_q(d) \geq \tau s_q(d_q^{(1)})\}|, 4, K), \quad (2)$$

where $s_q(d)$ is the post-splice, pre-deep-CE candidate-pool score, $d_q^{(1)}$ is the document at rank 1 of topic q , $\tau = 0.20$, and K is the maximum-depth

cap defined above. The threshold determines the topic-specific submission depth k_q ; after deep-tail reranking, the system submits the top- k_q prefix of the final ranking. The anchor is the rank-1 score rather than the pool maximum. Because the spliced ranking can contain entries whose scores originate from different fusion computations, s_q is not necessarily monotone in rank, and the maximum-scoring document can occur below the protected splice boundary. Using the maximum instead of rank 1 changes k_q for 88 of the 119 official topics. The cutoff is computed before deep-tail reranking. The deep stage leaves k_q unchanged but can alter both the order and membership of the submitted prefix.

4.2 Bounded Agentic Evidence Acquisition

Figure 2 shows the independent RAG pipeline, Bounded Agentic Evidence Acquisition with Citation-Aware Dense Writing. It starts from a BM25 candidate set, reads a bounded initial set, and asks a sufficiency judge whether to issue valid follow-up queries and read more evidence. Starting at BM25 depth 1,000, the initial 12-document read is the first acquisition iteration (iteration 0 in logs). Each later iteration reads up to 10 unseen documents, for at most six total iterations and 60 successfully read documents.

The initial ranking, as well as every fused ranking produced after a valid follow-up acquisition iteration, is passed through a top-300 reranking request, handled by an auxiliary inference service. Among the incoming top 300, a MiniLM cross-encoder scores successfully fetched text truncated to 256 words. For scored documents, the cross-encoder order is reciprocal-rank fused 1:1 with the incoming BM25/RRF-derived order:

$$s(d) = \sum_{r \in \{\text{incoming}, \text{CE}\}} \frac{1}{60 + \text{rank}_r(d)}. \quad (3)$$

Fusion retains the incoming ranking’s lexical prior instead of replacing it with the cross-encoder order. No stored evaluation supports a numeric pure-cross-encoder ablation for this component. Documents whose text could not be fetched are appended after the fused head, before the remaining, unreranked incoming-order tail. Passage reading requests at most three passages per document and falls back to document-level reading, capped at 200 lines, when the passage service is unavailable. The RAG checklist is generated offline from the narrative, independently of the Retrieval facet expansion in §4.1, and

supplied as evidence-coverage guidance. It does not use organizer-provided nugget annotations or relevance judgments. Its generated `vital/okay` checklist priorities are runtime guidance, not the evaluator’s vital-nugget labels used in V_{strict} . Runtime sufficiency, follow-up, dense-answer, and verify/revise calls use `gpt-5.6-sol` at temperature 0. Separately, `gpt-oss-120b` generates the offline checklist and judges nugget coverage and citation support in the public RAGDoll development evaluator; it is not the answer generator. Exact prompt-version identifiers remain in submission metadata and Appendix E.

The sufficiency judge returns an `enough` decision or candidate follow-ups. Deterministically valid proposals are searched at depth 1,000 as accumulated BM25 routes and fused with the persistent anchor BM25 route (weights 0.25 each and 1.0, respectively; RRF $k = 60$). The loop stops on sufficient evidence, an exhausted document or iteration budget, no valid follow-up after validation, or an unrecovered judge failure; Appendix C gives the validator rules.

The dense-writing branch constructs sentence-level answer objects and a reference list from the bounded evidence set. The finalization order is dense writing, trim to the implementation cap, verify/revise, a second trim, citation reordering, and strict JSONL validation. The 900–1,010-word dense-writer instruction is a generation target, not a validated final lower bound: verify/revise can rewrite or remove sentences, and the final validator enforces the upper limit but no minimum. Verify/revise works in batches of 20 sentences when batching is needed and adopts a revised answer only if strict validation succeeds and the result is non-empty; otherwise it returns the original answer. Citation reorder uses sentence-evidence scores when available. If the evidence-scoring service is unavailable, the original citation order is retained. The implementation trim cap is 1,020 words, below the organizer-specified maximum answer length.

5 Experiments and Results

Experimental protocol. All effectiveness experiments and system comparisons reported in this section use the 22-topic development set; the official 119-topic data are used only for structural submission diagnostics. Retrieval uses three projected UMBRELA qrels, relevance threshold $\text{rel} \geq 2$,

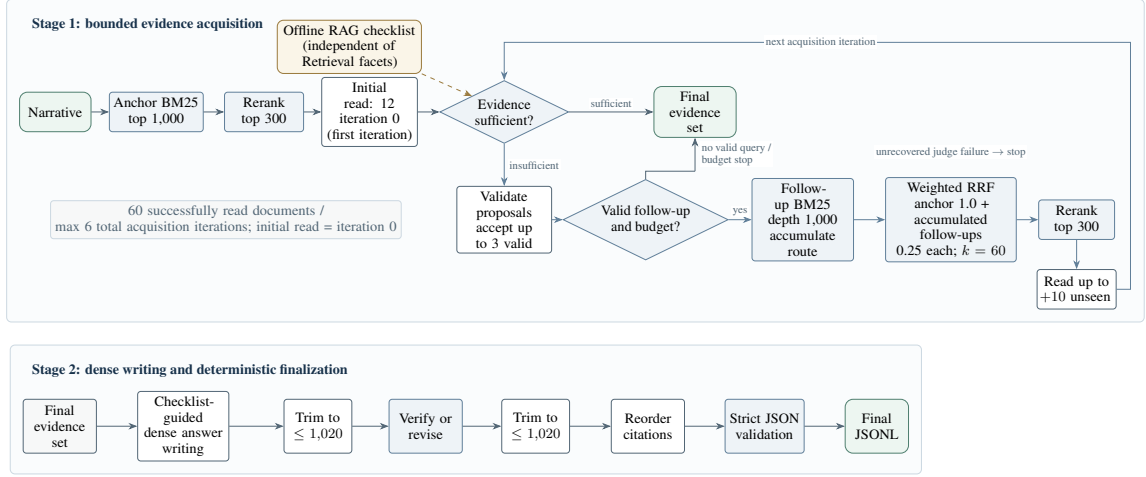


Figure 2: Bounded Agentic Evidence Acquisition with Citation-Aware Dense Writing. The checklist is a supplied artifact; the loop is bounded by reading and evidence-acquisition-iteration limits before dense writing and citation finalization. The checklist guides both the sufficiency assessment and the structure of dense answer generation. The recorded acquisition limits (12 initial/up to 10 per iteration/60 total/at most 6 iterations) are identical between the development and official 119-topic runs. The supplied checklists contain 7–9 facets per development topic (median 8) and 6–10 per official topic (median 9). Dense-writer evidence limits for the test-set submission configuration are reported separately in Appendix A.

and linear-gain nDCG. P@10, Recall, MAP, and MRR use binarized $\text{rel} \geq 2$ judgments. For each Retrieval run, we evaluate independently against the three projected UMBRELA qrels and report the arithmetic mean of the resulting aggregate metrics. The qrels are not merged. We validate rank continuity, duplicate identifiers, JSON structure, citation indices, and response-length compliance before submission.

Official submission. Our team (team ID 2026 cfda rag) submitted official runs for the 119-topic test set. Team submission records list cfda-vfs-deep (priority 1), corresponding to the selected deep-tail retrieval configuration, and cfda-vfs-unc (priority 2), corresponding to the selected pre-deep-tail retrieval configuration described in §4.1, as the two Retrieval runs. They list cfda-w5c (priority 1) as the RAG run, corresponding to the selected bounded 60-document acquisition configuration described in §4.2. Both Retrieval submissions were serialized with the variable-depth rule of Eq. 2 at $K = 5,000$, giving per-topic depths from 137 to 4,218 (median 1,080); each run file contains 176,224 rows. The two runs share the same underlying candidate pool, unchanged top-100 prefix, and per-topic depth. Deep-tail reranking changes the tail order for every topic and, for 103 of 119 topics, changes which documents fall inside the submitted prefix. At $K = 1,000$, an intermediate official serialization

made 63 of 119 topics hit the cap. Because the task specifies no maximum depth and the pool held up to 5,000 documents, we retained the development-selected $\tau = 0.20$ and rank-1 anchor and raised only the team cap to $K = 5,000$, reducing cap domination. This decision used structural diagnostics, not official judgments. Official test-set scores are assigned by the track organizers after submission and are not yet available at the time of this draft; all scores reported below are development-set diagnostics, as noted throughout.

The selected Retrieval output is the variable-depth system. The fixed top-1000 output is retained as a fixed-depth comparison because it uses a different submission depth: its Recall@1000 of 0.3159 is not substituted for the variable-depth Recall@submitted- k of 0.2343.

For RAG, we compare the 12-document-cap baseline, the 60-document deep-reading configuration, and the selected bounded 60-document acquisition configuration; extended reading is reserved for failure analysis. For V_{strict} , the selected-minus-deep-reading mean paired topic-level difference is -0.0042 , with an approximate 95% paired t interval of $[-0.0497, +0.0412]$ and an 8/11/3 win/loss/tie record. Selection reflects stronger citation-support behavior; the V_{strict} difference is inconclusive, and the stored variants do not support a single-factor causal interpretation.

Key implementation and operating parameters

for both systems are summarized in Appendix Table 4 (Retrieval) and Table 5 (RAG).

5.1 Retrieval Results

Table 1 reports the stage-wise Retrieval evidence. The pool columns are not interchangeable with the submitted variable-depth measure. Table 2 isolates the cutoff change from the later deep-tail reranking.

The Retrieval output obtains $P@10=0.9076$ while submitting fewer than 1,000 documents for 20 of the 22 narratives. Its mean depth is 601.45, median depth 687.5, and the 18 distinct depths show that the output is not padded. The fixed-depth comparison is $\text{Recall}@1000=0.3159$; it is not substituted for $\text{Recall}@submitted-k$.

The $n\text{DCG}@10$ increase from 0.7414 to 0.7456 is caused by the cutoff change rather than by deep-tail reranking. It arises because the historical CE-threshold cutoff returned only four documents for topic 515, whereas the relative cutoff exposes ranks 5–10 from the same pre-deep ranking. That historical cutoff uses sigmoid-transformed cross-encoder logits stored in the top-100 fusion artifact. It emits four documents when no artifact entry reaches 0.50 (the archived `ce_dead` flag); otherwise it counts entries with sigmoid score at least 0.50 (the archived `ce_calibrated` field), clips that count to [4, 5,000], and takes the corresponding candidate-pool prefix. Its observed development depths are 4–94. The variable- k scores are computed from the post-splice, pre-deep-CE candidate-pool scores; Deep CE changes ordering but not the topic-specific submission depth.

The selected pre-Deep-CE pool has pool $\text{Recall}@1000=0.2937$, pool $\text{Recall}@5000=0.4895$, and $\text{Recall}@submitted-k=0.2204$. Deep-tail reranking preserves that pool while changing the corresponding values to 0.3159, 0.4895, and 0.2343, respectively. In this same-pool comparison, deep-tail reranking improves pool $\text{Recall}@1000$ without changing pool membership.

5.2 RAG Results

The RAG analysis compares stored dense-writing configurations under different evidence-acquisition and writing settings. Table 3 reports the selected development diagnostics for the three stored RAG configurations without the internal PS diagnostic; the unreported remainder of FS and NS is PS.

The 12-document-cap baseline reads five initial documents and up to four unseen documents per acquisition iteration, with a 12-document cap

and at most three acquisition iterations. Both 60-document configurations record 12 initial documents, up to ten unseen documents per acquisition iteration, a 60-document acquisition cap, and at most six acquisition iterations, plus reranking, passage selection, a supplied checklist, and verify/revise. The archived deep-reading launcher sets a 40-document writer limit; the selected development record preserves neither writer-document nor character limits. Official limits appear in Appendix A. The rows are therefore stored system variants, not a controlled writer-visible-evidence ablation.

The selected RAG output contains 22 valid and completed topics. Each topic has 12–30 references, and each answer contains 44–71 answer objects. The mean answer length is 976.73 words and the maximum is 1,019 words. Table 6 in Appendix D reports the corresponding structural statistics for both systems side by side.

Figure 3 places the two main comparisons on their relevant scales. With the 5,000-document Retrieval pool fixed, deep-tail ordering raises pool $\text{Recall}@1000$ by 0.0222 and $\text{Recall}@submitted-k$ by 0.0139. Figure 3 also summarizes the RAG coverage–support trade-off.

5.3 Answers to the Research Questions

RQ1. The selected multi-route pool reaches pool $\text{Recall}@1000$ of 0.2937, compared with 0.2635 for original-narrative BM25. The original narrative remains an explicit retrieval route, while the protected splice retains the top 200 entries of the multi-route, head-reranked pre-facet ranking before facet-derived candidates are added to the complementary tail. Deep-tail ordering further raises the same-pool $\text{Recall}@1000$ to 0.3159.

RQ2. Variable-depth Retrieval produces topic-specific submission depths of 100–1,000 documents, with a mean depth of 601.45 and $\text{Recall}@submitted-k$ of 0.2343; 20 of 22 development topics end below 1,000 documents. The selected RAG configuration bounds evidence acquisition at 60 documents and at most six evidence-acquisition iterations and obtains an all-nugget score of 0.5298. The stored RAG configurations show an observed coverage–support trade-off. We do not claim a compute-efficiency advantage for variable-depth Retrieval because runtime cost was not evaluated under a controlled comparison.

RQ3. The 60-document deep-reading configuration has higher all-nugget coverage (0.5485

Table 1: Representative Retrieval configurations on the 22-topic development set. The rows are not a strictly cumulative ablation: the pool columns and the submitted-depth column refer to different objects, and only the final two rows are a clean same-pool ordering comparison. Boldface marks a column maximum, not statistical significance.

Configuration	Pool $R@1000$	Pool $R@5000$	Submitted $R@k_q$
Original-narrative BM25	0.2635	—*	—
Query-expanded retrieval with head reranking	0.2945	0.5155	—
Facet-expanded retrieval with protected head	0.3005	0.4938	—
Selected pre-deep-tail retrieval configuration	0.2937	0.4895	0.2204
Selected deep-tail retrieval configuration	0.3159	0.4895	0.2343

*The original-narrative baseline was retrieved only to depth 1,000.

Table 2: Variable-depth cutoff comparison on the 22-topic development set; all values are development diagnostics, not official test-set results. nDCG uses linear gains after grades below 2 are mapped to zero; Recall is always Recall@submitted- k in this table. Boldface marks a column maximum, not statistical significance.

Cutoff stage	Submitted depth	nDCG@10	Recall@submitted- k	MAP	P@10	MRR
Historical CE-threshold cutoff	4–94	0.7414	0.0563	0.0514	0.8985	0.9848
Relative $\tau = 0.20$	100–1,000	0.7456	0.2204	0.1389	0.9076	0.9848
Deep-tail reranking (same depth)	100–1,000	0.7456	0.2343	0.1432	0.9076	0.9848

Table 3: Selected RAG development diagnostics for three stored RAG configurations. These are system variants, not a single-factor ablation. Boldface marks a column maximum, not statistical significance.

Configuration	V_{strict}	A	FS	NS
12-document-cap baseline	0.4145	0.4862	94.8%	0.1%
60-document deep-reading	0.4698	0.5485	95.0%	0.1%
Bounded 60-document acquisition (selected)	0.4656	0.5298	97.0%	0.1%

versus 0.5298), whereas the selected bounded 60-document acquisition configuration has higher Full Support (97.0% versus 95.0%). This comparison shows the coverage–citation-support trade-off observed across the two configurations, not a Pareto improvement. Because the writer, verifier, and evidence budgets were not isolated in a single-factor ablation, the stronger citation support is a property of the selected stored configuration rather than an identified causal effect of verification.

5.4 Development Variants and Failure Analysis

Query2Doc introduces pseudo-document vocabulary while the original narrative remains an anchor. Facet routes expose separately searchable requirements. To prevent facet-derived candidates from reordering the pre-facet head, we retain the pre-

facet top-200 prefix and incorporate facet-derived candidates only into the complementary tail.

Recall@submitted- k depends on the topic-specific submitted cutoff and must therefore be evaluated on the submitted file. The development Retrieval output spans 100–1,000 documents per topic; the corresponding official 119-topic outputs span 137–4,218. The fixed top-1000 development output provides a fixed-depth comparison with Recall@1000=0.3159. The deep CE tail preserves the 100-document head and changes tail ordering through the specified fusion with pool order; the topic depth remains determined by post-splice, pre-deep-CE candidate-pool scores.

The 12-document-cap baseline is descriptively lower than both 60-document configurations in Table 3. The selected bounded 60-document acquisition and 60-document deep-reading configurations form the close pair; their V_{strict} difference is inconclusive, and selection reflects stronger citation-support behavior rather than a statistically significant V_{strict} advantage. A larger-budget extended-reading configuration encountered server errors on most generation calls, while its recorded failure status did not reliably surface those failures. Because execution was incomplete and failure reporting was unreliable, we exclude it from formal comparison in Table 3.

The final word-limit cap enforces the 1,024-word output contract, which the selected RAG output sat-

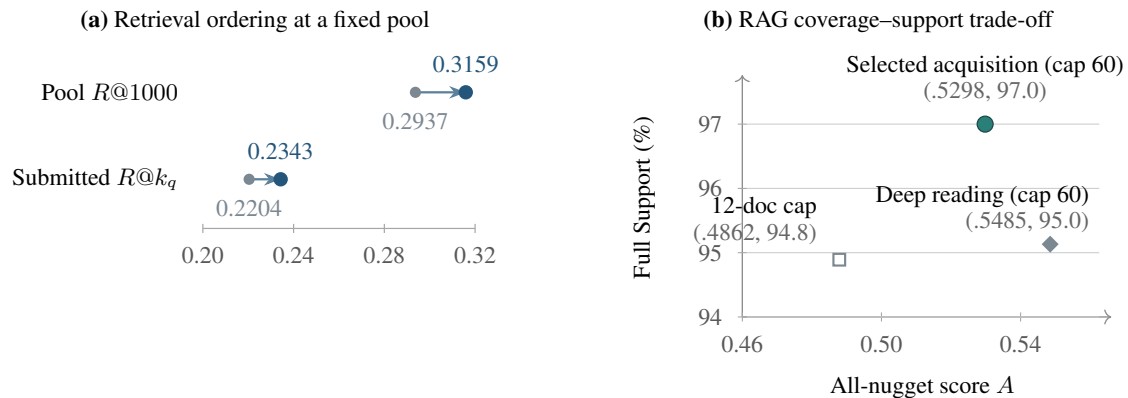


Figure 3: Development-set comparison of the selected systems and their nearest development alternatives. Panel (a) reports the observed change from pre-deep-tail to deep-tail ordering on the same 5,000-document Retrieval pool. Panel (b) plots RAG all-nugget coverage against Full Support; it illustrates the observed coverage–citation–support trade-off. Exact values and metric definitions appear in Tables 2 and 3.

isfies; it does not establish that trimming improves answer semantics.

6 Limitations

Effectiveness is limited to the 22-topic development set; organizer test scores were unavailable, and no statistical superiority is claimed. Missing writer-visible parameters limit single-factor RAG interpretation. Immutable provider model/index snapshots and a versioned top-head service checkpoint are not retained. The original source used to produce the official deep-tail run is not retained, although a reconstructed replay reproduces the frozen output byte-for-byte. Changes to an auxiliary RAG inference service during official execution prevent exact per-topic service-version attribution. Priorities rely on team metadata because no organizer receipt is retained.

7 Conclusion

We developed separate Retrieval and RAG pipelines for TREC RAG 2026. The Retrieval results show that a protected head can be combined with deep-tail reranking while the variable-depth cutoff remains tied to post-splice, pre-deep-CE candidate-pool scores. The RAG results show that coverage and citation support must be examined together when comparing stored RAG configurations.

Both pipelines separate protected or bounded evidence-selection stages from later ranking and answer-construction stages: Retrieval protects the head before tail reranking, and RAG caps additional reading before dense writing and verification.

Further experiments should compare adaptive stopping policies under controlled reading budgets. Organizer judgments, when released, will be needed to determine whether the development-set patterns transfer to the test topics.

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A Parameters

A.1 Retrieval parameters

The selected Retrieval configuration uses the original-narrative route, Query2Doc, separate follow-up queries, weighted RRF, a 5,000-document candidate pool, top-100 BM25/cross-encoder/dense reranking, facet-derived tail expansion, protected top-200 merge, deep-tail reranking, and threshold-derived variable depth.

Table 4: Key Retrieval operating parameters and observed output depths.

Parameter	Value
Original-query repetitions in Query2Doc route	5
Initial evidence read	5 documents
Follow-up queries per acquisition iteration / RRF weight per accepted BM25 route	1–3 / 0.25
New evidence per acquisition iteration	Up to 4 unseen documents
Evidence-acquisition cap	12 documents
Maximum acquisition iterations	3
Candidate pool depth	5,000
Head rerank depth	100
Protected pre-facet prefix	200
Deep CE tail (development)	Ranks 101–5,000; CE/pool RRF 1:1, $k = 60$
Deep CE tail (official)	Ranks 101–3,000; ranks 3,001–5,000 retain pool order
Variable-depth threshold	0.20
Minimum depth clip	4
Development maximum K	1,000
Official maximum K	5,000
Observed development output depth	100–1,000
Observed official output depth	137–4,218

A.2 RAG parameters

Table 5: RAG evidence-loop and generation parameters. Loop parameters apply to the selected bounded 60-document configuration in development and official runs; dense-writer limits are from the official submission configuration.

Parameter	Value
Initial BM25 depth	1,000
Initial documents read	12
New documents per acquisition iteration	Up to 10 unseen documents
Maximum documents read	60
Maximum acquisition iterations	6
Passages per document	3
Document read limit	200 lines
Top-300 rerank boundary	300
Anchor / follow-up RRF weights; k	1.0 / 0.25 each; 60
Checklist	Supplied artifact
Verify/revise batch	20 sentences
Verify/revise mode	Weaken unsupported or over-extended claims
Dense-writer evidence (official)	Up to 60 documents; 1,200 characters each
Dense-writer target (generation instruction)	900–1,010 words; not a final lower bound
Runtime sufficiency/query model	gpt-5.6-sol, temperature 0
Dense writer and verify/revise model	gpt-5.6-sol, temperature 0
Offline checklist generator	gpt-oss-120b, temperature 0
Development RAGDoll judge	gpt-oss-120b
Implementation trim cap	1,020 words
Organizer-specified maximum answer length	1,024 words

The retained source identifiers are Xenova/ms-marco-MiniLM-L-6-v2 for the top-head cross-encoder, bge-m3 for the dense endpoint, and [cross-encoder/ms-marco-MiniLM-L6-v2](#) for the selected deep-tail cross-encoder. The frozen run records do not preserve a versioned provider checkpoint for the top-head service, so no more specific head checkpoint is asserted here. For Retrieval, Query2Doc and facet-query generation use gpt-5.6-sol at temperature 0. Query2Doc has one 400-token call and omits its route after failure or unusable output; facet templates are schema-validated, with up to four attempts. The Retrieval sufficiency/follow-up judge uses gpt-oss-120b at temperature 0, with up to five total attempts. Table 5 separately summarizes RAG runtime model roles, and its prompts appear in Appendix E. This table is a concise summary;

stage-specific fusion and per-topic trace details are specified in the Method and reproducibility records.

B Additional Development Variants

The fixed top-1000 comparison illustrates the depth trade-off but is not the selected variable-depth Retrieval output. The 12-document-cap baseline and the two 60-document configurations are development comparisons. The extended-reading configuration is included for failure analysis.

Variable-depth metric computation. The deep-tail reranking Recall@submitted- k figure (0.2343, Table 2) is computed by applying the variable-depth rule in Equation 2 to the post-splice, pre-deep-CE candidate-pool scores.

C Follow-up Query Constraints and Validation

The validator lowercases and extracts `[a-z0-9-]+` tokens, requiring 4–12. It rejects a normalized duplicate of any prior query (including the anchor) or earlier accepted proposal in the iteration; any `?`, brace, period, semicolon, colon, or triple-backtick fence; and tokens equal to: obtain, find, source, sources, detail, detailed, comprehensive, concrete, examples, evidence, information, overview, history, impact, provide, directly, addresses, address, account, explain, or the token *and*. It also rejects an anchor-overlap ratio above 0.75, computed as the number of query-token occurrences whose token also appears in the anchor divided by the number of distinct tokens in their union. At most the first three valid proposals are retained. A parse-invalid judge response triggers retries for up to five total attempts; a parsed insufficient response with no valid proposal stops the loop with no valid follow-up query.

D Structural Statistics

The following table records comparable output statistics; contract checks are described in the main text.

Table 6: Selected development-output structure. Four-number sequences denote minimum / mean / median / maximum.

Property	Retrieval	RAG
Topics	22	22
Output records	13,232 total rows	44–71 objects/topic
Depth / refs per topic	$k_q \in [100, 1,000]$ docs	12–30 refs
Mean / median	601.45 / 687.5 docs	22.41 / 22 refs
Distinct depths	18	–
Word count	–	723 / 976.73 / 1,001 / 1,019
Citations per object	–	1 / 1.0288 / 1 / 3

E Prompts and Execution Traces

The official 119-topic RAG launcher used three LLM prompt templates in the main runtime path: a combined sufficiency-and-query prompt, a dense answer writer, and a sentence-level verify/revise pass. Citation ordering and final structural validation are deterministic and use no additional LLM prompt. The submission metadata stores the base version `agentic_rag_baseline_v1`; the writer template identifies itself more specifically with the `-dense` suffix. The templates below are transcribed from the runtime implementation. Only typesetting has been normalized; runtime-injected content is omitted.

The template text is reproduced verbatim; some heuristic instructions and examples are not guaranteed to satisfy every stated constraint simultaneously.

The combined sufficiency/query template is embedded in the iteration runner, while the writer and verifier are shared prompt-module templates. The offline checklist consumed by the sufficiency judge (§4.2) is produced separately, once per topic, by a single system/user decomposition call (retried on parse failure, temperature 0). Stable source mappings, file hashes, and replay locations are recorded in the project reproducibility record rather than exposed as machine-specific paths in the paper.

Offline checklist generation (system and user prompt). This prompt runs once per topic, before the agentic loop, to decompose the narrative into the sub-aspects the checklist enumerates. Its sub-narrative decomposition produces the format consumed by the sufficiency judge and dense writer. It does not use organizer-provided nugget annotations or relevance judgments.

System: You decompose a research narrative into the distinct sub-aspects

a complete answer must cover. You return strict JSON only - no prose, no markdown fences.

User:

Narrative:

[NARRATIVE]

Decompose this narrative into the distinct sub-aspects a complete, well-rounded answer must cover. Return strict JSON with exactly this shape:

```
{ "sub_aspects": [
  { "title": "<short noun-phrase name of the aspect, 3-8 words>",
    "importance": "vital" | "okay",
    "expected_facts": ["<one-sentence fact a good answer would state>", ...] }
]}
```

Rules:

- 5 to 12 sub_aspects, mutually distinct (no overlapping or duplicate aspects).
- "vital" = the narrative explicitly asks about it or an answer omitting it would clearly be incomplete; "okay" = secondary/supporting aspect.
- 2 to 4 expected_facts per aspect, each a single short declarative sentence.
- Titles must be specific to this narrative, not generic ("Economic impact of X on Y", not "Impacts").
- JSON only. No text before or after.

The resulting sub-aspects are mapped to checklist items by a deterministic checklist-conversion step. The conversion rule is checked against the 22-topic development checklist before applying it to the 119 test topics.

Evidence sufficiency and

follow-up queries. Return ONLY one JSON object. No markdown. No explanation.

Required schema:

```
{ "enough": boolean, "queries": string[] }
```

If the evidence already supports the topic, return

```
{ "enough": true, "queries": [] }.
```

If evidence is insufficient, return { "enough": false, "queries": [...] } with 1-3 BM25 keyword queries.

Query rules: 4-12 English tokens;

keyword phrase, not sentence/question; one aspect only; no duplicates.

Forbidden words: obtain, find, source, sources, detail, detailed, comprehensive, concrete, examples, overview, history, impact, provide, explain.

Previous queries: [PREVIOUS QUERIES]

Topic: [NARRATIVE]

Aspect checklist (use when judging sufficiency and choosing follow-up queries; prefer queries that target aspects not yet covered by the evidence):

[CHECKLIST ITEMS]

Evidence:

[DOCUMENT IDS AND EXCERPTS]

After an empty model response, the runtime varies the final retry line to avoid replaying a

cached empty response. The retry suffix does not change the task definition.

Dense answer generation.

Prompt version: agentic_rag_baseline.v1-dense
You generate an evidence-grounded TREC RAG answer.

Use only the provided evidence documents. Do not use outside knowledge.

LENGTH: the complete answer MUST total between 900 and 1010 words. Short answers waste the official budget and lose coverage. Keep adding supported factual sentences until you approach 1010 words.

SENTENCE STYLE - every sentence must be:

- atomic: exactly one factual claim per sentence, at most ~35 words;

- concrete: preserve named entities, numbers, dates, organizations, places verbatim from the evidence;

- grounded: cited to the evidence documents that state the fact.

- maximally SPECIFIC: prefer the most specific fact available - named people, organizations, places, causes and their outcomes, findings and their evidence. A precise claim (who did what, why, with what result) scores; a general statement about the same aspect does not.

Do NOT write generic overview sentences that only describe the topic in the abstract (e.g. 'This is an important and dynamic field', 'X is part of culture'). Such sentences are worthless and must be omitted.

Good: "Persistent gender and racial pay gaps affect athlete compensation, and business interests shape both media-rights deals and team management."

Bad: "Athlete compensation is a controversial and important topic."

No introduction, no conclusion, no hedging, no restating the question - only supported facts.

Organize the answer by these aspects of the topic; write roughly 80-120 words for EACH aspect (a few sentences each), in this order:

[CHECKLIST ITEMS]

Citations are zero-indexed positions into the references array, not docids. Each sentence may cite at most three references.

References must be unique ClimbMix docids and every reference must be cited at least once.

Do not include uncited retrieved documents in references.

Return only strict JSON. Do not use Markdown fences, comments, or explanatory prose.

Required JSON shape:

```
{ "references": ["shard.00000.00000"],
```

```
  "answer": [{ "text": "Supported
```

```
    sentence.", "citations": [0] } ] }
```

```
[TOPIC ID AND NARRATIVE]
```

Evidence documents:

[UP TO 60 DOCUMENTS,

1,200 CHARACTERS EACH]

Sentence-level verify/revise (production mode: weaken).

You are revising a cited RAG answer so that every sentence is fully supported by its cited evidence.

For each sentence below, compare the sentence against its cited evidence excerpts and act:

- Fully supported: keep the sentence unchanged with the same citations.
- Over-extended (partially supported): rewrite it so it claims ONLY what the excerpts support. Do not add new claims.
- Not supported by its citations: rewrite it into a weaker related claim that the excerpts DO support (hedge, narrow, or generalize it); if other cited excerpts support the point, switch its citations to those; remove the sentence only when nothing in any excerpt relates to it.

Rules: do not invent new references or citations; keep citation indices pointing into the references array below; each sentence at most three citations; keep the overall answer coherent and under 1024 words; keep the original sentence order for kept/rewritten sentences.

Return only strict JSON, no Markdown fences, exactly this shape:

```
{"references":["shard.00000.00000"],
"answer":[{"text":"Supported sentence.", "citations":[0]}]}
```

The references array must stay the same list (unused entries are allowed and will be pruned automatically).

[TOPIC ID AND NARRATIVE]

References array (index -> docid):

[REFERENCES]

Sentences to verify:

[BATCH OF AT MOST 20 SENTENCES, CITATIONS, AND EVIDENCE EXCERPTS]

Citation ordering and final validation (no LLM prompt).

After verify/revise, the runtime sends each answer sentence and its cited document IDs to the evidence-scoring service with a 200-character evidence budget. It sorts multiple citations by the returned evidence score. If the service is unavailable, it leaves the original order unchanged. Citation reordering therefore has no LLM prompt to reproduce.

Worked trace: development topic 144. The narrative asks for a comparison of why banks such as Silicon Valley Bank fail, the hidden risks to depositors, and how banks differ from credit unions. The sufficiency judge ran for three iterations before stopping:

- **Iteration 0** (12 documents read):

enough=false; follow-up queries requested on FDIC/NCUA regulatory oversight, credit-union community lending, and depositor/investor risk exposure.

- **Iteration 1** (10 more documents read): enough=false; follow-up queries requested on ownership/governance differences and service/fee comparisons.
- **Iteration 2:** enough=true; loop stops with stop_reason=enough after 3 iterations total.

The summary is drawn from the complete stored development record for topic 144: its judge trace, iteration trace, and per-topic execution status records. It is a development example and is not taken from the separately archived official 119-topic run. Exact artifact identifiers and hashes are provided in the project reproducibility record.

Stopping Behavior on the 119 Test Topics. All 119 official topics completed. Table 7 aggregates the per-topic execution status records: most topics exhaust the reading budget rather than stopping because the judge reports sufficiency, and most reach the 6-iteration cap. The caps were active constraints, not fallback limits: 82 topics stopped at the document budget and 89 reached iteration 6.

Table 7: Stop reason and iteration count across all 119 official topics, aggregated from per-topic execution status records.

Stop reason	Topics	Share
Maximum documents read (60)	82	68.9%
Judge reports sufficient (enough)	36	30.3%
No valid follow-up query	1	0.8%

Iterations used	Topics
2	5
3	8
4	4
5	13
6 (cap)	89

Mean iterations were 5.45 (median 6, cap 6). Documents read successfully ranged from 22 to 60 (mean 55.04, median 60, cap 60); no topic failed outright. All 119 topics completed without runtime failures.